Final Project:
Amplified Speaker with Power Supply

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Lab Section: 33
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1. Introduction

The purpose of this experiment was for the student to apply everything they have learned this year in Engineering Electronics lecture and in the lab and apply it to building an iPod docking station. The iPod docking station was to be designed as an inexpensive one that could be sold during the holiday season. The student at minimum had to employ a 2 stage amplifier design and was recommended to use the Darlington Configuration to achieve the output power requirement.

2. Background

A Class AB Output Stage is when a biasing voltage is when a bias voltage VBB is applied between the bases of multiple transistors (for the circuit below it would be Q1- Q3), creating an increase in a biasing current which in turn eliminates crossover distortion. Crossover distortion in audio power amplifiers (which is what this project is constructing) creates unpleasant sounds.

A Darlington Pair configuration is a variation of the CE-CC circuit where the collector of one transistor is connected to another transistor whose collector is connected to that of another transistor. The Darlington Pair configuration is used to implement a high-performing voltage follower.

In this Final Project, you were tasked with building a music amplifier with a power a power supply that followed the following specifications:

Specifications:

• Power supply: 120 VRMS at 60 Hz.
• Input Signal (vin, not Vsig) = 350mVRMS, frequency: entire audio frequency range • Load: 8Ω speaker
•OutputPower: **10**W RMS +/-10%
• Volume Control Required
• LED indicator for Power Supply

Requirements:

• At minimum, a 2-stage amplifier design must be employed

• A Darlington Configuration for the output stage is suggested to achieve output power requirement

3. *Materials and Equipments*

Components:

- (2) 50Ω resistors, (2) 15 kΩ resistors, (1) 20 kΩ resistor, (3) 1k kΩ resistor, (1) 400 Ω resistor, (2) 100 Ω resistor, (2) 500 Ω resistor
- (2) 470 µF Capacitors, (3) 330 µF Capacitors
- (1) 2N3904 NPN Transistor, (3) TIP31AG NPN Transistor, (1) TIP32AG PNP Transistor
- (9) 1N4002G zener diodes
- (1) Voltage Regulator (LM7912CT)
- (1) Voltage Regulator (LM7812CT)
- (2) LED Diodes
- (1) Transformer (1P2S)

Equipments:

- Function Generator
- DC Power Supply
- Digital Multimeter (DMM)
- Oscilloscope
- BNC Cables
- Banana to Alligator Test leads
- Jumper Wires
- (2) Breadboards

4. *Experimental Procedures*

4.1 - Initial Calculations

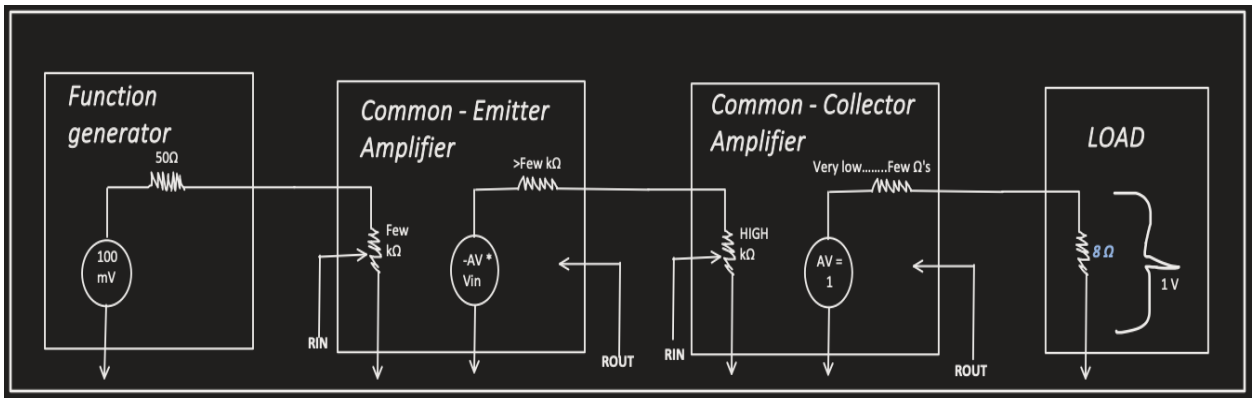


Figure 1.1- Block Diagram

Final design project

Designing Common-Emmitter Amplifier :-

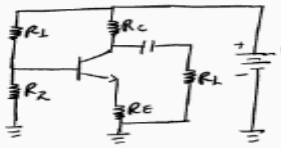
1) The input impedance (RIN) of the common-collector amplifier enables us to determine the Rout of the common-emitter amplifier.

Thus, using rule of thumb, If Rout (of CE amp) = 1/10 RIN (of cc amp), then

Rout = 1/10 (14kΩ) = 1400 Ω = 1.4 kΩ, we need Vpeak of 12.649 V at the load. Since the 1.4k will drop about 10% of the voltage, we make our CE amp supply about 12.649 V + 10% = 13.9139 since input V is 494.97475 mVpeak, we need

$A_V = \frac{V_{out}}{V_{in}} = \frac{13.9139}{494.97475m} = 28.11$ V/V gain

Given Circuit below



So, we know that Rout ≈ Rc = 1.4kΩ

Setting $V_C = \frac{1}{2}V_{CC}$, => $V_C = \frac{1}{2}(24) = 12$ V

using ohm's law, $I_C = \frac{(V_{CC}-V_C)}{R_C} = \frac{(24-12)}{1.4k} = 8.57142$ mA

the gain of non-bypass CE amp is: $A_V = -R_C/R_E$

which is $R_E = \frac{-R_C}{A_V}$ => $R_E = \frac{-1.4k\Omega}{-28.11} = 49.80434\ \Omega$

Since $I_E \approx I_C$, $I_E = 8.57142$mA, then $V_E = I_E \cdot R_E = 8.57mA \cdot 49.8 = 0.426894$ V

$V_B = V_E + 0.7V = 1.126894$ V, then $V_{CE} = V_C - V_E = 12 - 0.426894 = 11.573$ V

from IV-Curve of 2N3904, we can determine that for $V_{CE} = 11.573$ V, if we want $I_C = 8.57142$mA, we need $I_B = 4.1 \times 10^{-5} = 41\mu A$,

Then, $\beta = \frac{I_C}{I_B} = \frac{8.6357mA}{41\mu A} = 210.6268$ A/A, then $R_{IB} = \beta r_e + (\beta+1)(R_E)$

thus, $R_{IB} = 634.1463 + (210.6268+1)(49.80434) = 10539.9331 = 10.5k\Omega$

=> rule of thumb $R_B = (1/10)(10.5k\Omega) = 1053.99331\ \Omega$

=> $V_{BB} = I_B * R_B + V_B = (41\mu A)\cdot(1053.99331) + 1.126894 = 1.17011$ V

=> $R_1 = \frac{V_{CC} R_B}{V_{BB}} = 21618.3431\ \Omega$ or $21.6k\Omega$ $\quad R_2 = \frac{V_{BB} R_1}{V_{CC} - V_{BB}} = 1108.014075\ \Omega$ or $1.1k\Omega$

=> $R_{in} = R_B \| R_{ib} = 958.17573\ \Omega$ $\quad A_V = \frac{I_{out}}{I_{in}} = \frac{I_{out} = \frac{V_{out}}{R_{out}}}{} = \frac{13.9139}{1400} = 9.9385$mA

$I_{in} = \frac{V_{in}}{R_{in}} = \frac{494.97475mV}{958.17573} = 516.58\mu A$

$A = \frac{I_{out}}{I_{in}} = \frac{9.9385mA}{516.58\mu A} = 19.239$ A/A

Figure 1.2 - Calculations

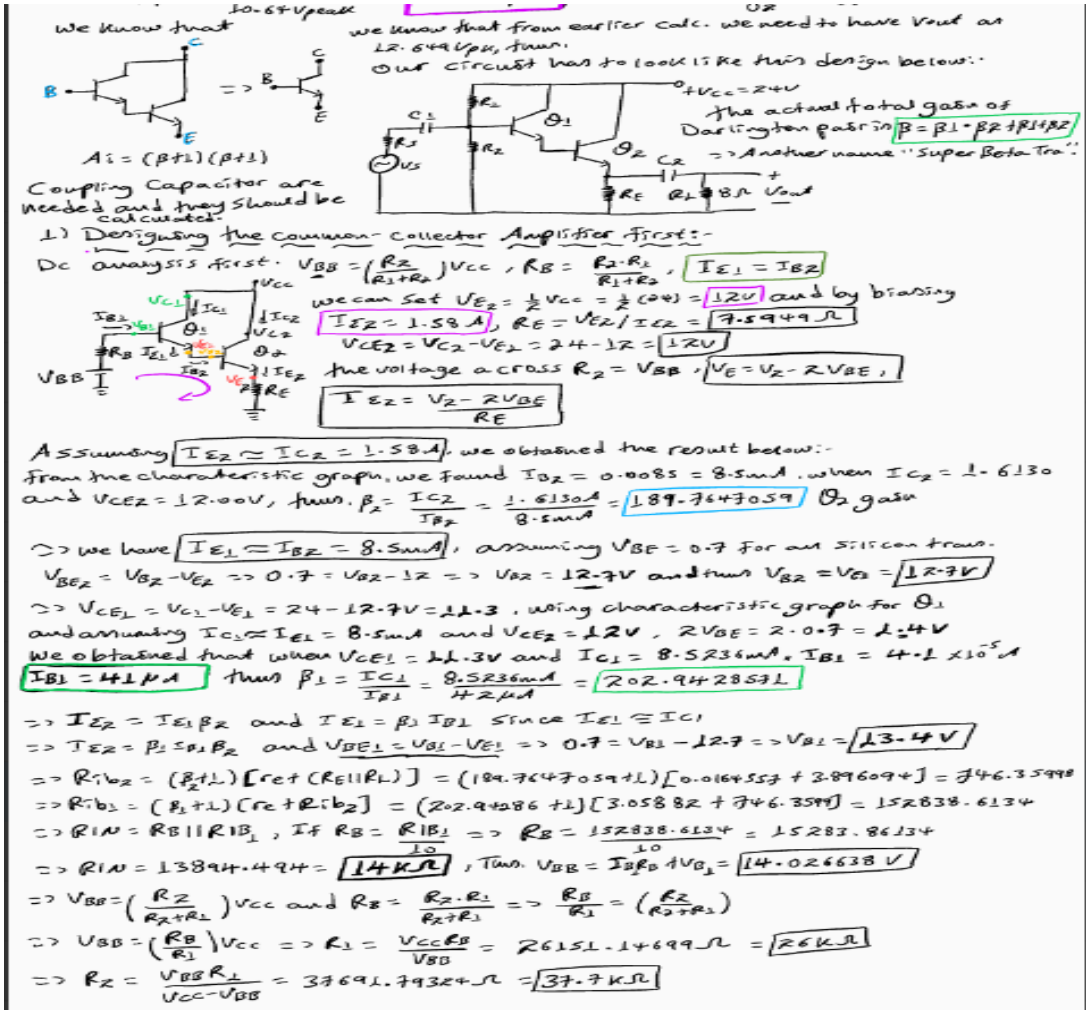


Figure 1.3 - Calculations

Compound darlington and NPN darlington was used to achieve the desired output voltage. The calculations were made based on the previous labs, and the input impedance of the CC was approximately 14kΩ.

4.2 - Design of Each Stage

CE stage was chosen because they are large-signal amplifiers with the objective of providing high power to the load. The CC stage that we used, which is the push-pull amplifier using Darlington transistors, can be used to increase the input resistance presented to the driving amplifier and avoid severely reducing the voltage gain. CC with one compound darlington stage was used to increase the performance of the overall CC amplifier.CC amplifier has little to no distortion.It is also known as impedance transformer or emitter follower.Compound pair

amplifier has higher output voltage, it's 99.5 % of the input voltage. This way the circuit has better thermal stability.

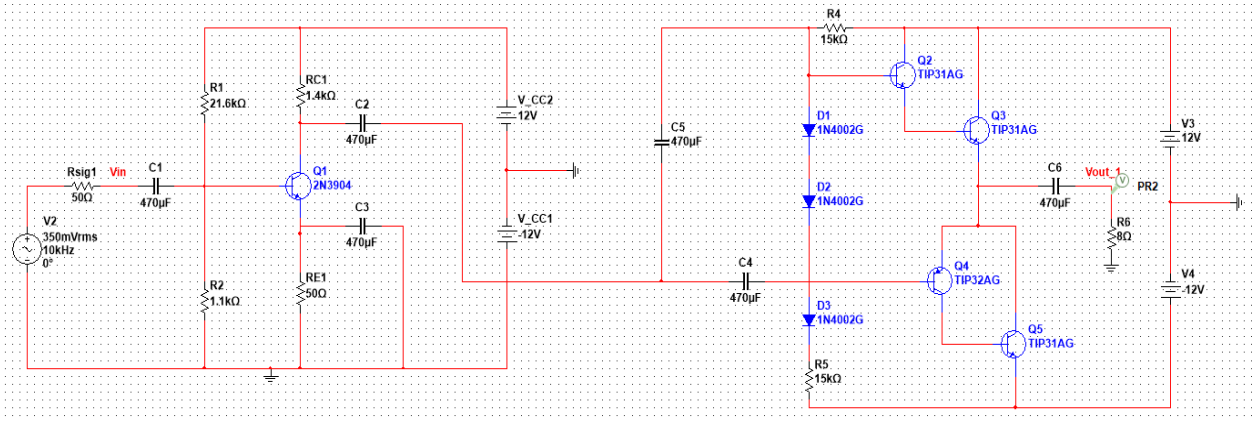


Figure 1.4- Cascaded circuit

The output of the CE is connected to the input of CC. The VCC was chosen 24 V and configured like this because in order to get at least an output voltage of 10V, the voltage across collector and emitter has to be half of 24 V, so that the ac output voltage can swing symmetrically across the center without being clipped.

4.3 Midterm Power Supply

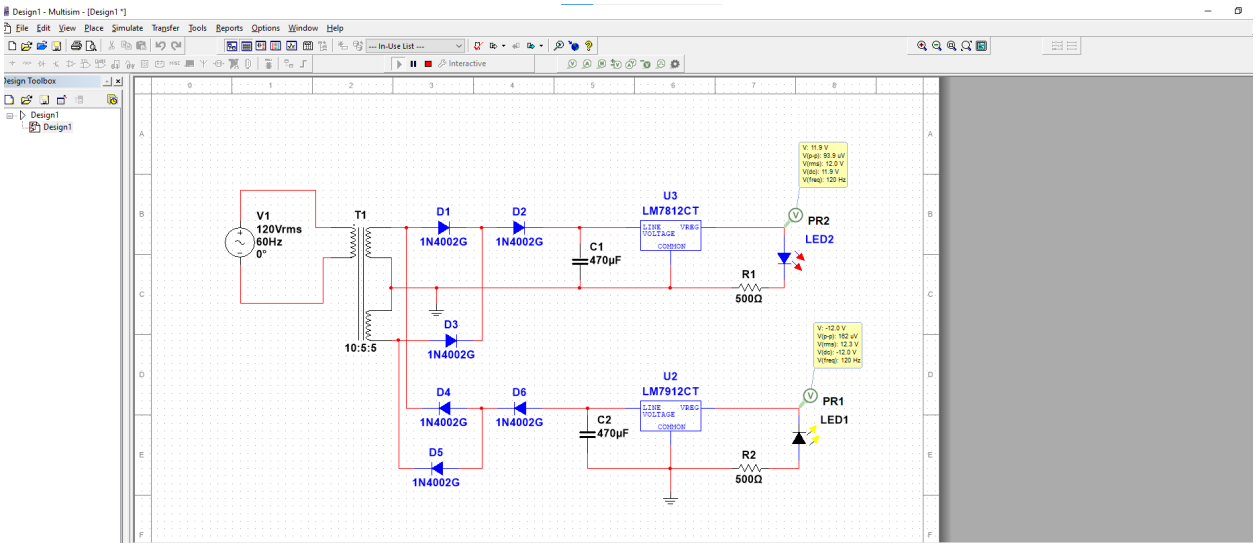


Figure 1.5- Power Supply

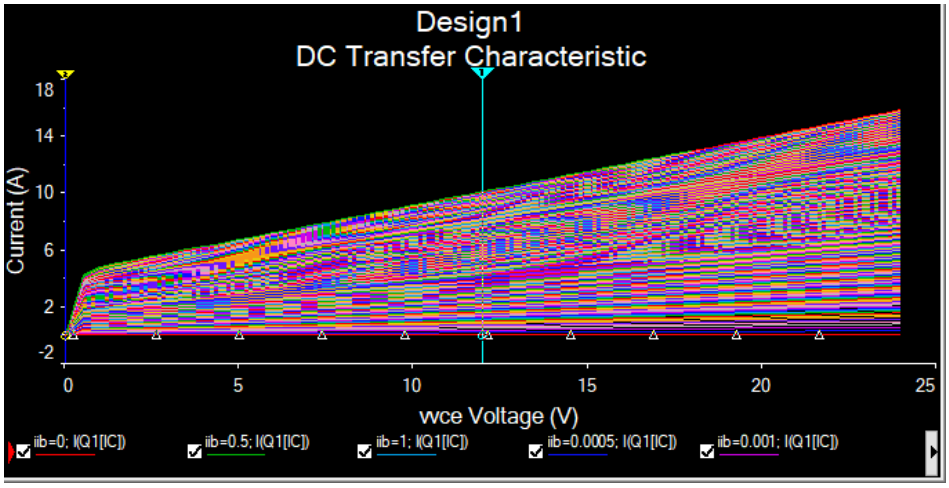
For the power supply, we used the exact design of our midterm project to support our final project design. The specifications of this power supply are as follows:

±12V AC-DC Power Supply

- ➢ The student cascaded the center-tapped transformer, full wave rectifier, a filter capacitor, a voltage regulator, and a LED Display to create an AC-to-DC power supply with the following specifications:
- ➢ Line Input Voltage: 120V RMS at 60Hz
- ➢ Two regulated Output Voltages: 12VDC and -12VDC o On State indicator (Red LED)
- ➢ Ripple <1%
- ➢ The students DC power supply can be seen in Figure
- ➢ The student measured the two output voltages of the circuit with the DMM to verify that it met the specifications of 12VDC and -12VDC. The student then used the oscilloscope to obtain needed measurements.

5. Measurements and Results

Determining Base current of Q2 (IB2):



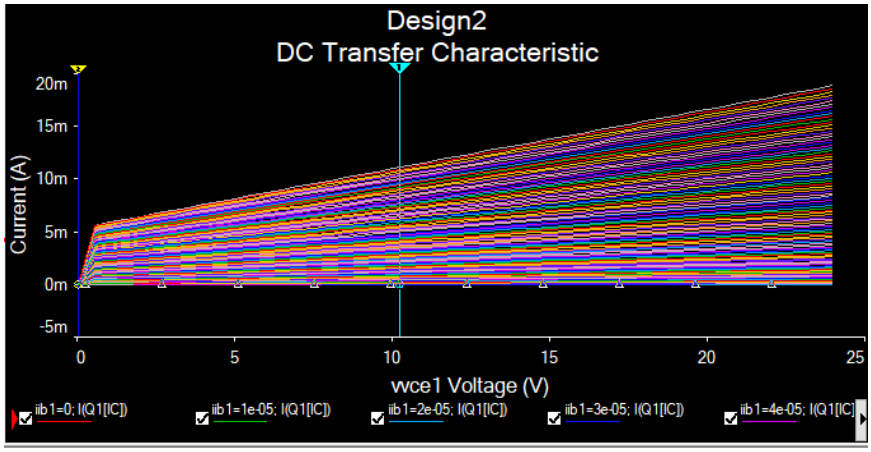

```
iib=0.0085; I(Q1[IC])

              11.9841
               1.6118
               0.0000
            305.4463µ
             -11.9841
              -1.6115
            134.4697m
```

```
iib=0.0085; I(Q1[IC])

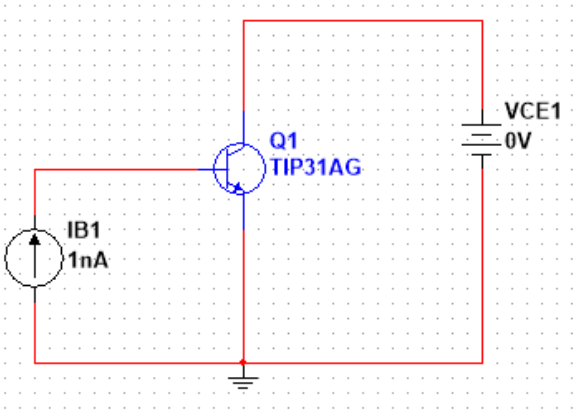
                12.0000
                 1.6130
                 0.0000
               305.4463µ
               -12.0000
                -1.6127
               134.3926m
```

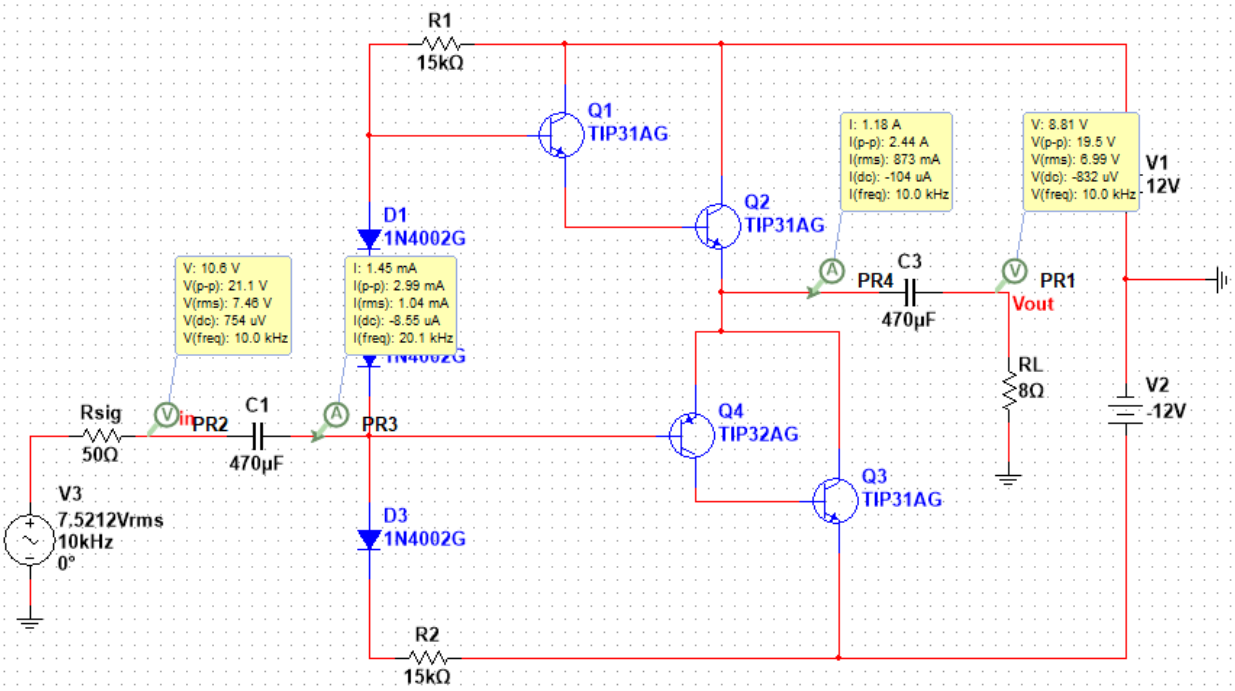
Determining Base current of Q2 (IB2):

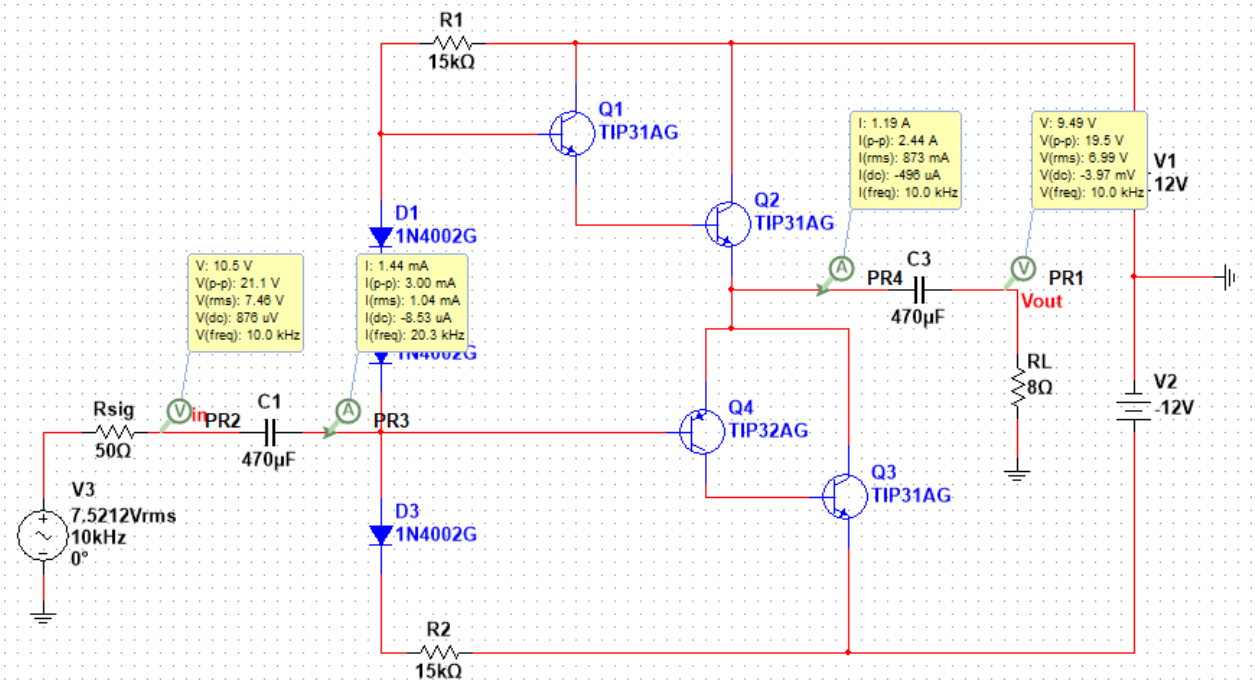



```
iib1=4.25e-05; I(Q1[IC])

                10.2000
                 8.5548m
                 0.0000
               -27.2810µ
               -10.2000
                -8.5820m
               841.3769µ
```

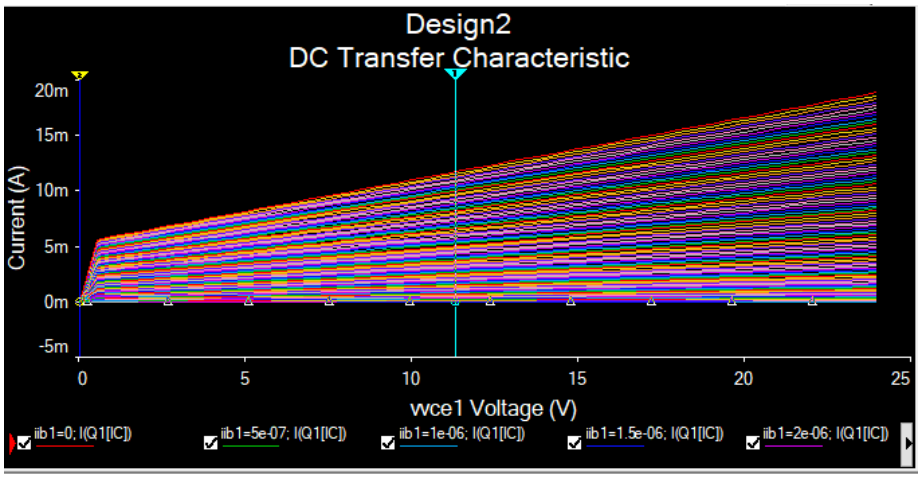

VCE1
0V
Q1
TIP31AG
IB1
1nA


R1
15kΩ
Q1
TIP31AG
D1
1N4002G
V: 10.6 V
V(p-p): 21.1 V
V(rms): 7.46 V
V(dc): 754 uV
V(freq): 10.0 kHz
I: 1.45 mA
I(p-p): 2.99 mA
I(rms): 1.04 mA
I(dc): -8.55 uA
I(freq): 20.1 kHz
Rsig
50Ω
PR2
C1
470µF
PR3
V3
7.5212Vrms
10kHz
0°
D3
1N4002G
R2
15kΩ
Q2
TIP31AG
I: 1.18 A
I(p-p): 2.44 A
I(rms): 873 mA
I(dc): -104 uA
I(freq): 10.0 kHz
PR4
C3
470µF
Q4
TIP32AG
Q3
TIP31AG
V: 8.81 V
V(p-p): 19.5 V
V(rms): 6.99 V
V(dc): -832 uV
V(freq): 10.0 kHz
PR1
Vout
RL
8Ω
V1
12V
V2
-12V



The real one:

Determining Base current of Q2 (IB1):



```
iib1=4.1e-05; I(Q1[IC])

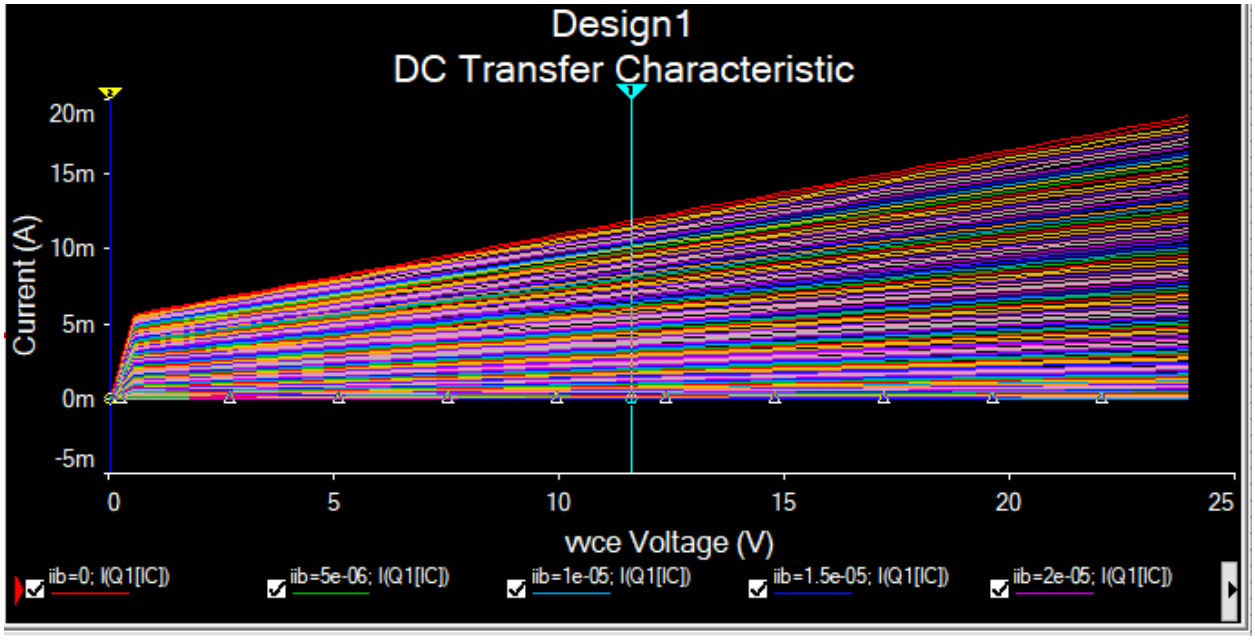
                  11.3000
                  8.5236m
                   0.0000
                 -26.4427µ
                  -11.3000
                  -8.5501m
                 756.6439µ
```

Table 1.1. Measurements of Current, Voltage, Power Gains

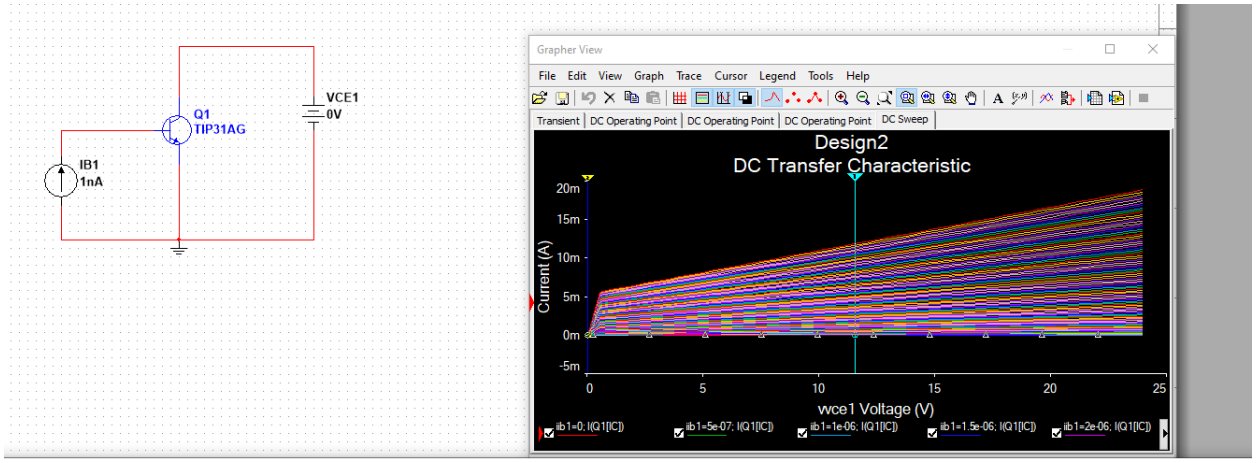
Values:	**Common - Emitter Amplifier**	***Common - Collector Amplifier***	***Entire Circuit Cascaded***	***Percent Error (Cal. vs. Sim.)***	***Percent Error (Cal. vs. Meas.)***	***Percent Error (Sim. vs. Meas.)***
Calculated Voltage Gain	23.503 V/V	0.9967 V/V	9.01 V/V	9.26%	3.94%	5.86%
Simulated Voltage gain	22.17 V/V	0.9867 V/V	9.93 V/V			

Measured Voltage Gain	23.21 V/V	0.9814 V/V	9.38V/V			
Calculated Current Gain	19.34 A/A	500 A/A	9670	10.87%	3.29%	15.89%
Simulated Current gain	19.239 A/A	563.98 A/A	10850.43 056			
Measured Current Gain	18.8A/A	498	9362.4			
Calculated Output Power	13W			3.17%	18.18%	14.54%
Simulated Output Power	12.6W					
Measured Output Power	11W					

Common Emitter Amplifier IV characteristics to find IB:



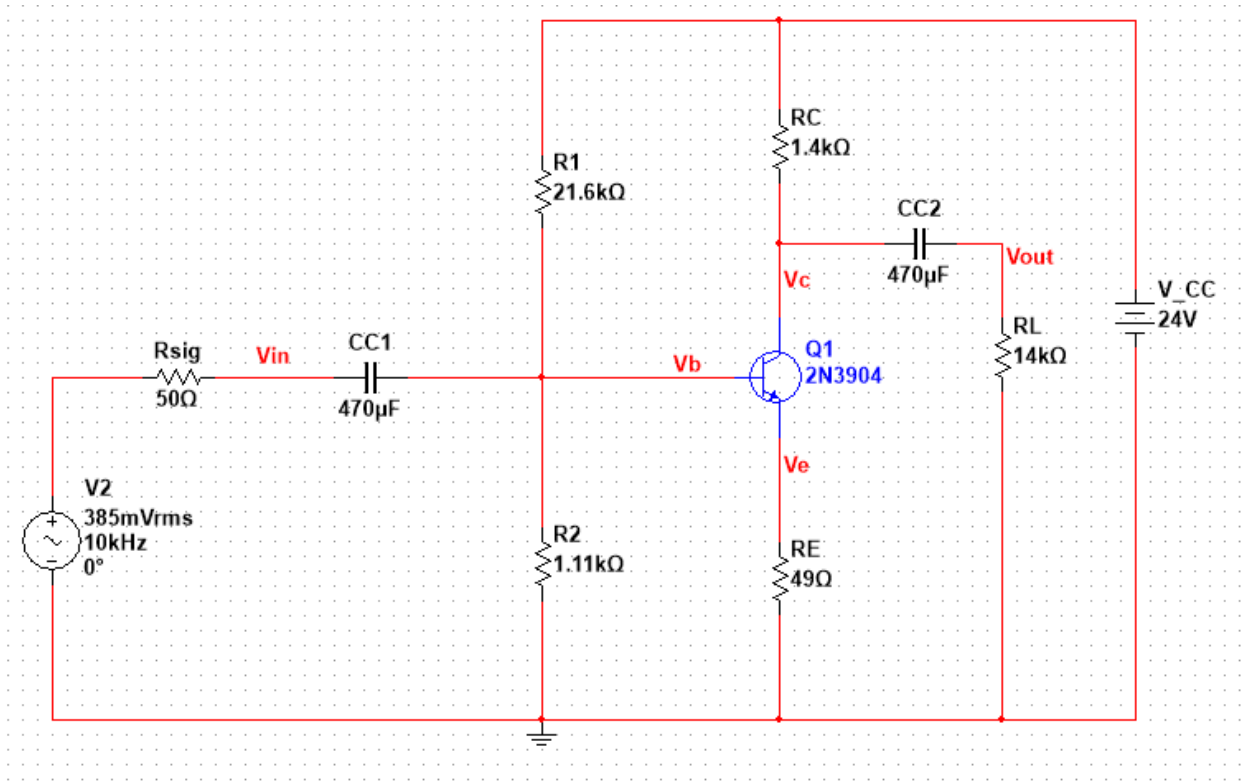
iib=4.1e-05; I(Q1[IC])
11.5730
8.6357m
0.0000
-26.4427µ
-11.5730
-8.6622m
748.4809µ



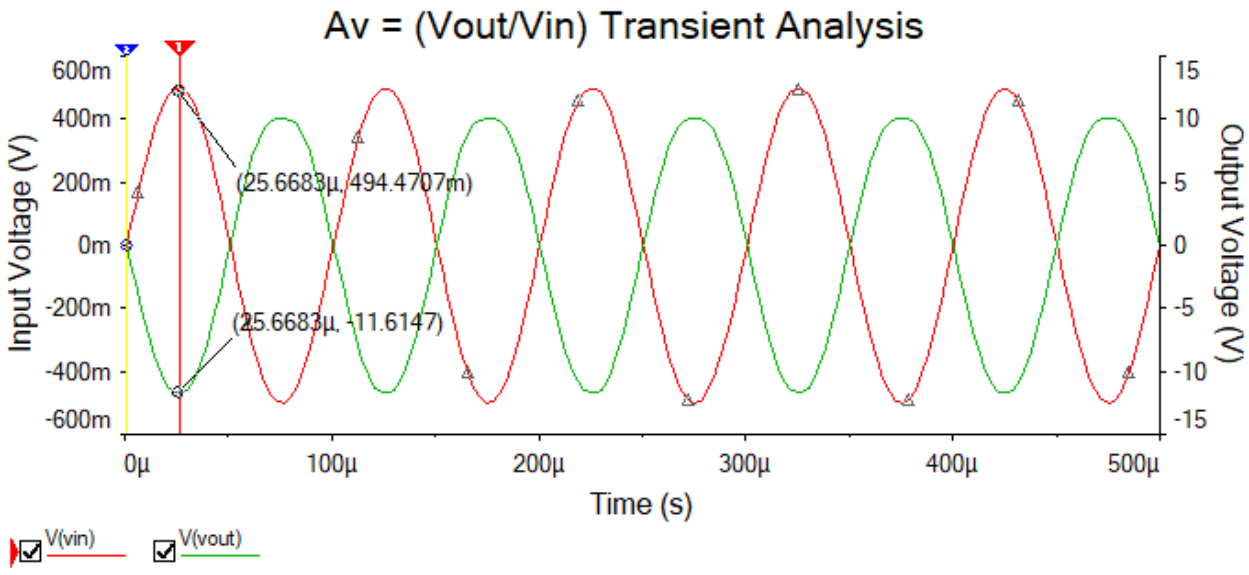
; I(Q1[IC]) iib1=4e-05; I(Q1[IC]) iib1=4.05e-05; I(Q1[IC]) iib1=4.1e-05; I(Q1[IC]) iib1=4.15e-05; I(Q1[IC]) iib1=4.2e-05; I(Q1[IC]) iib1=4.25e-05; I(Q1[IC]) iib1=4.3

11.5370	11.5370	11.5370	11.5370	11.5370	11.5370	11.5370
8.1138m	8.2819m	8.4509m	8.6209m	8.7919m	8.9638m	9.1366m
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
-25.5983µ	-25.8805µ	-26.1619µ	-26.4427µ	-26.7228µ	-27.0022µ	-27.2810µ
-11.5370	-11.5370	-11.5370	-11.5370	-11.5370	-11.5370	-11.5370
-8.1394m	-8.3078m	-8.4771m	-8.6474m	-8.8186m	-8.9908m	-9.1639m
705.5074µ	720.0982µ	734.7747µ	749.5352µ	764.3784µ	779.3027µ	794.3068µ

Circuit For CE -



Voltage Gain:



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	V(vin)	V(vout)
x1	25.6683µ	25.6683µ
y1	494.4707m	-11.6147

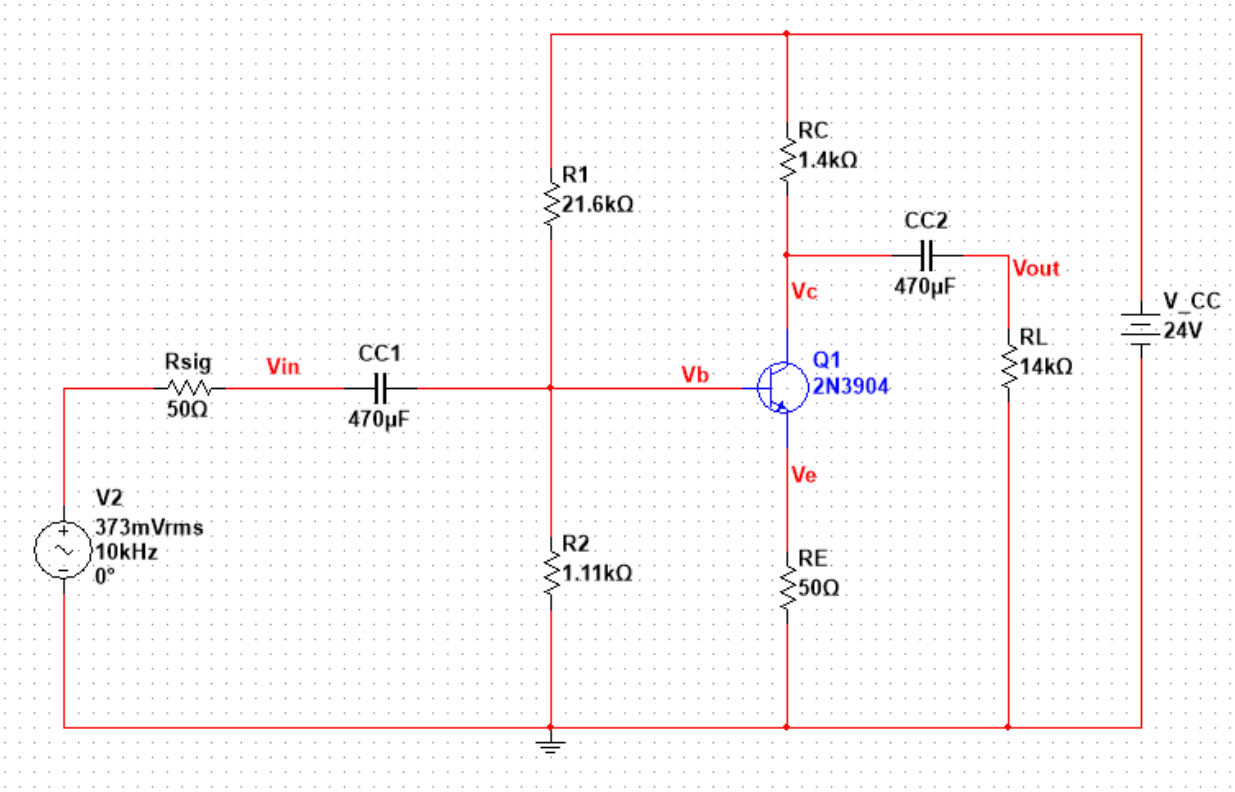


Figure 1.2. Simulation of common emitter amplifier

The completed circuit,

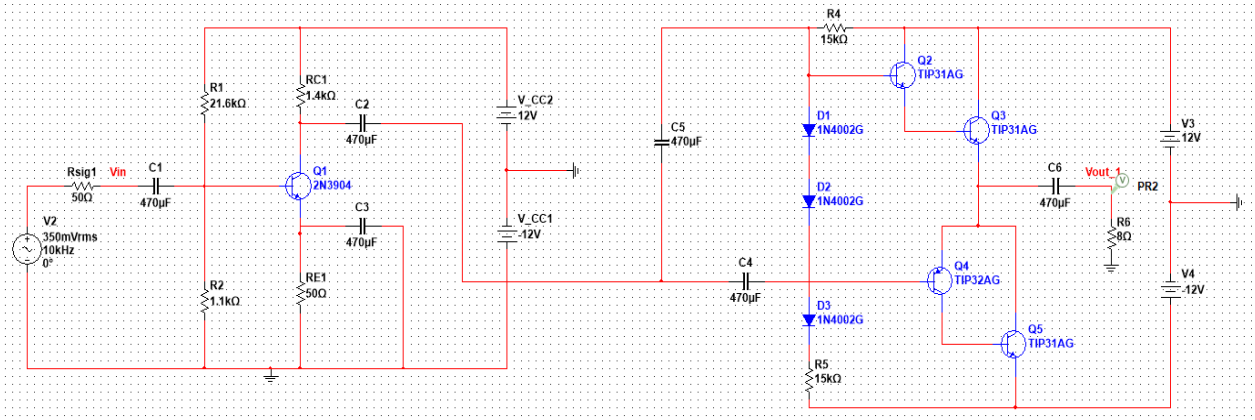
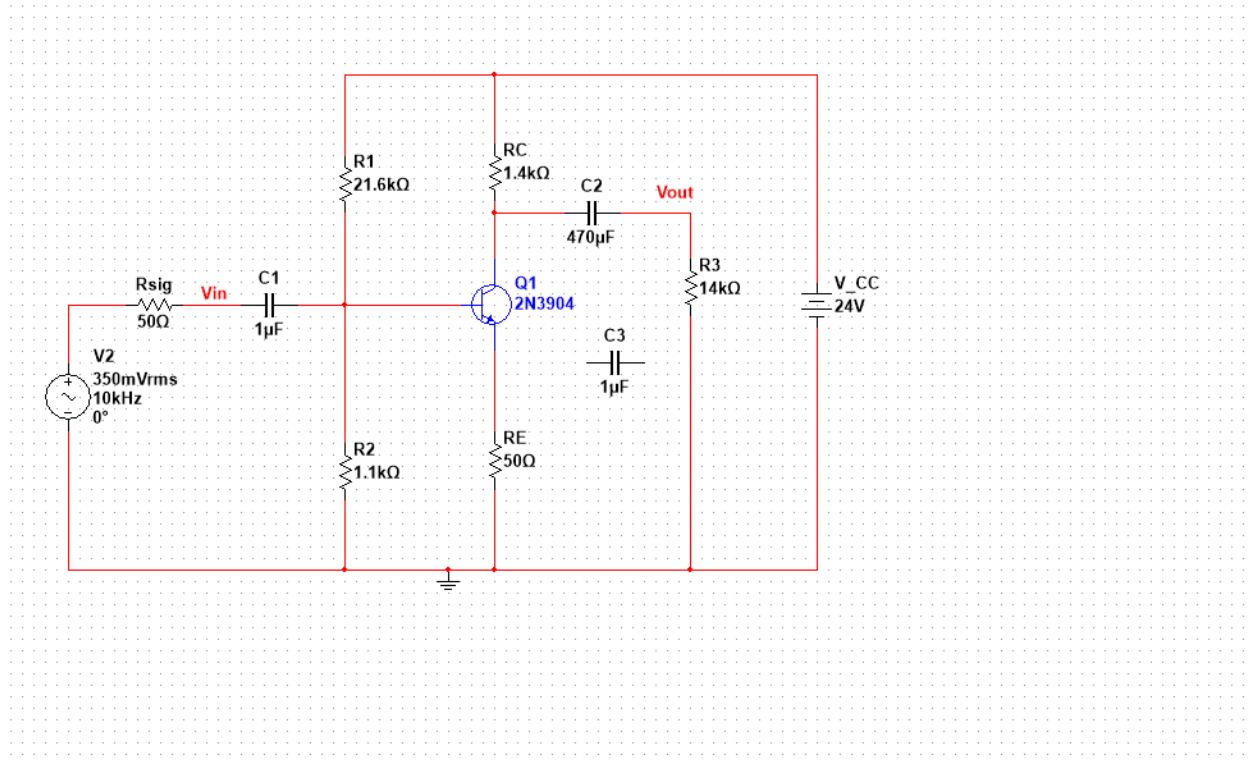
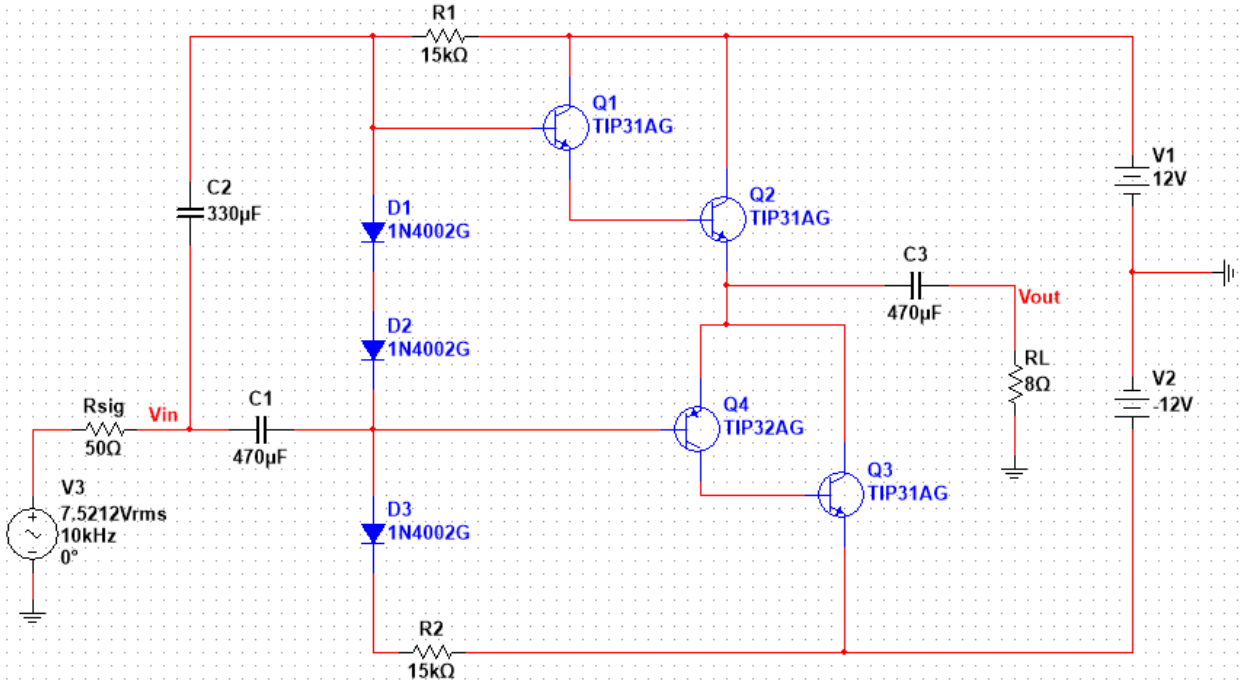


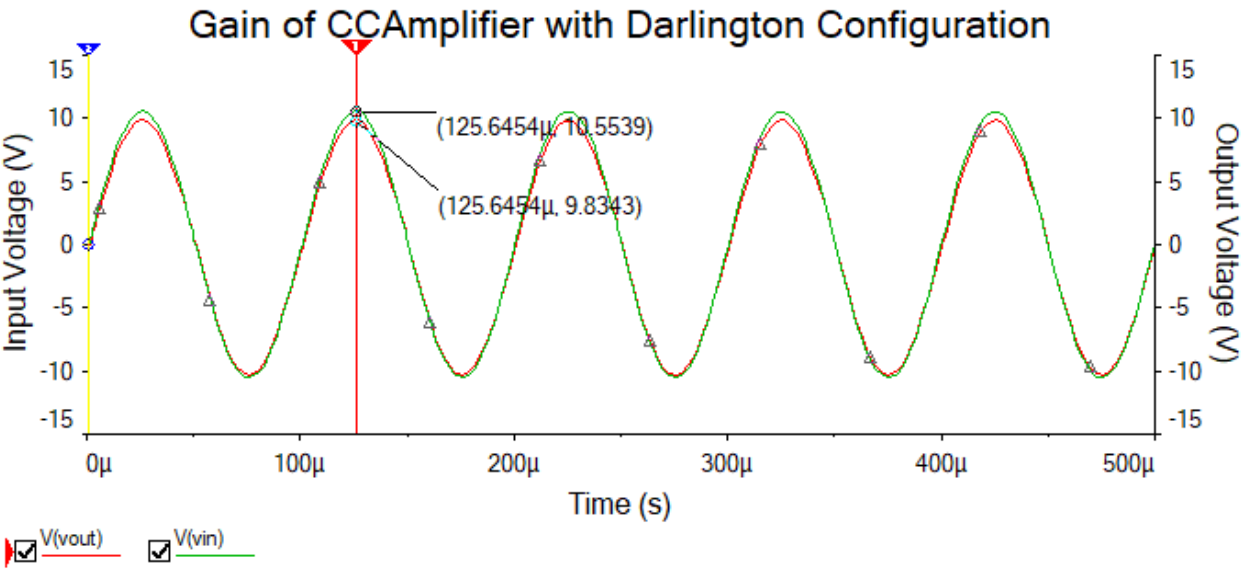
Figure 2.1. The entire cascaded circuit



Simulation Results:

1.

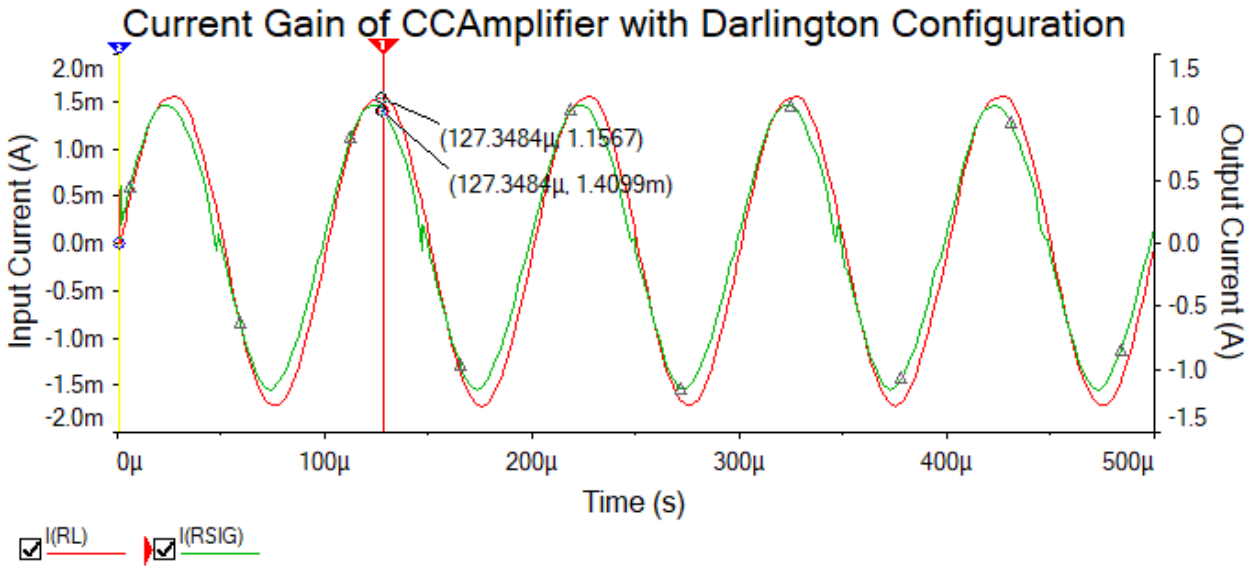




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	V(vout)	V(vin)
x1	125.6454µ	125.6454µ
y1	9.8343	10.5539

2. Current Gain

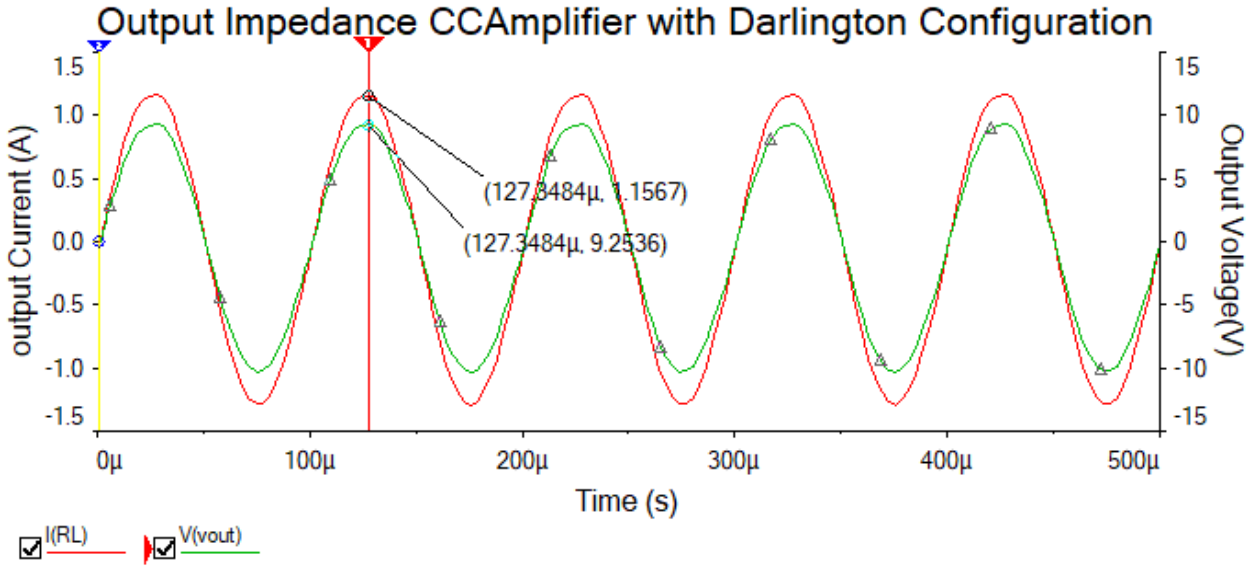


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	I(RL)	I(RSIG)
x1	127.3484µ	127.3484µ
y1	1.1567	1.4099m

Figure 2.1. Current gain of the Second Stage

3. Output impedance



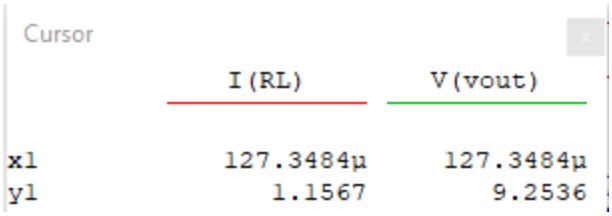


Figure 3.1. Output impedance of Second Stage

4. Input Impedance

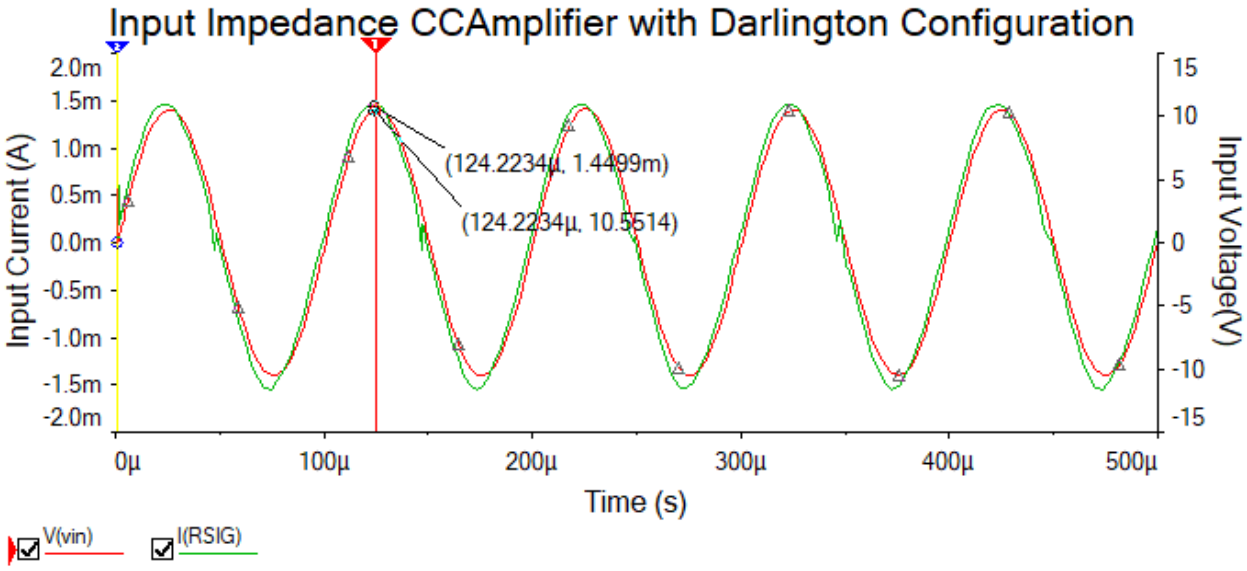


Figure 4.1. Input impedance of Second Stage



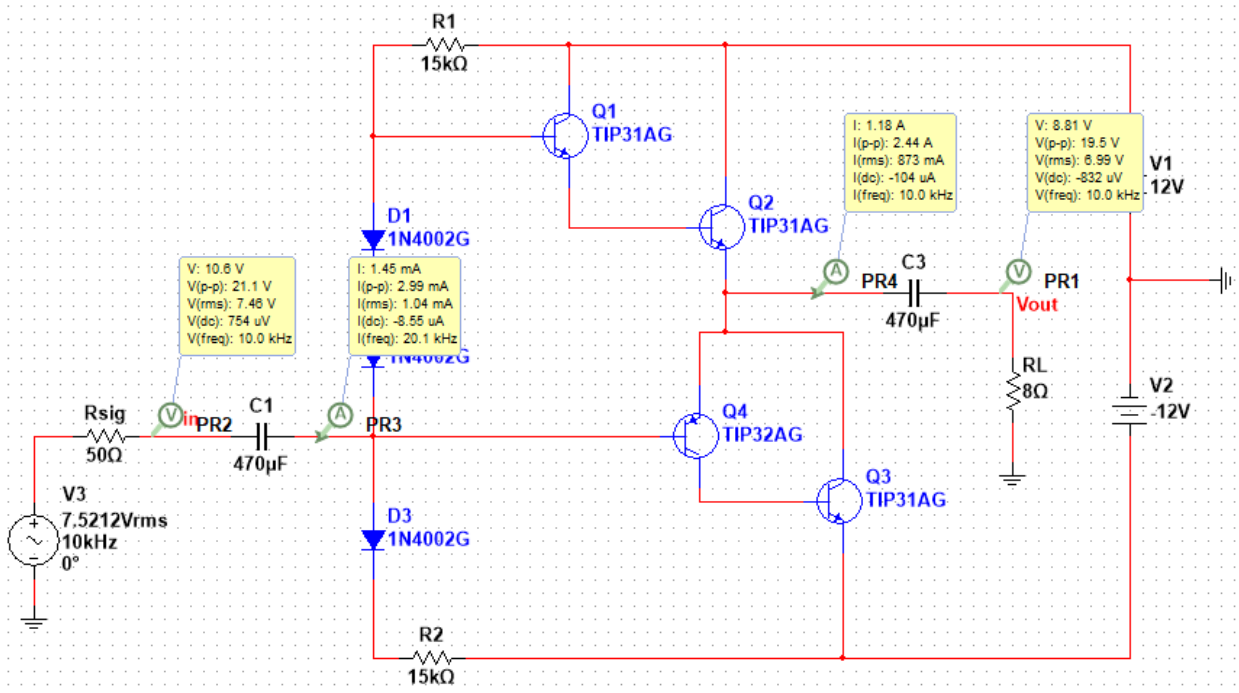


Figure 4.1. The second stage CE amplifier simulation circuit

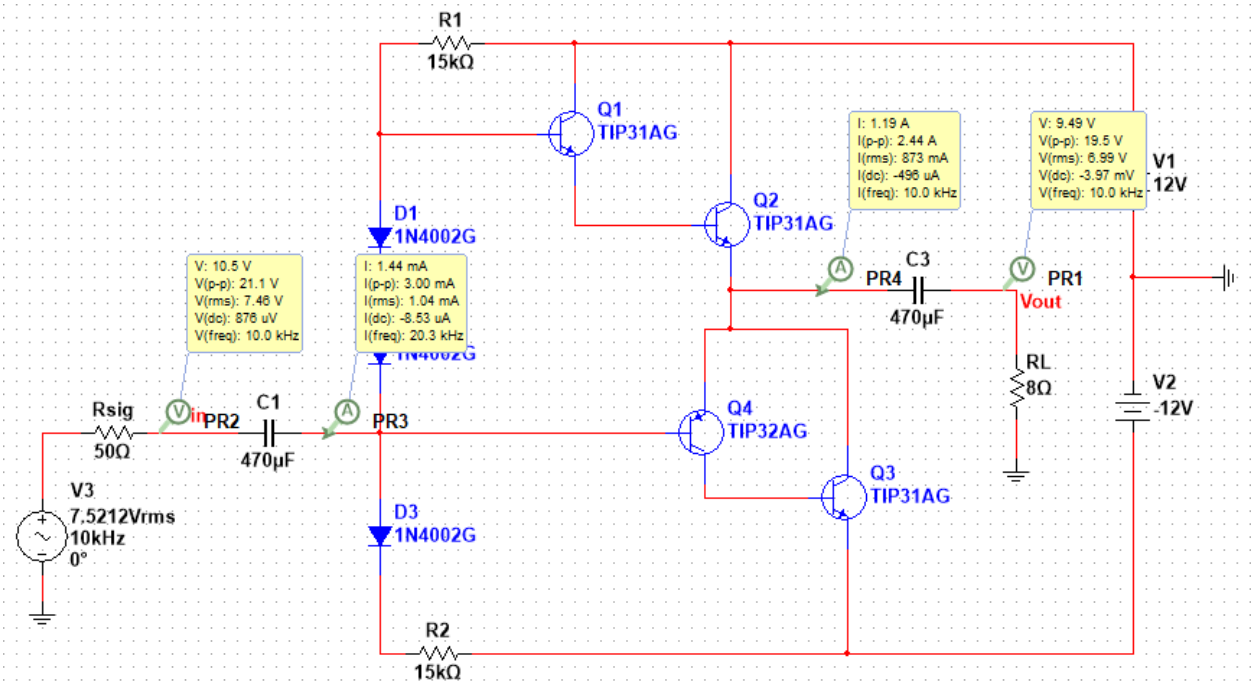


Figure 4.2. The second stage CE amplifier simulation circuit

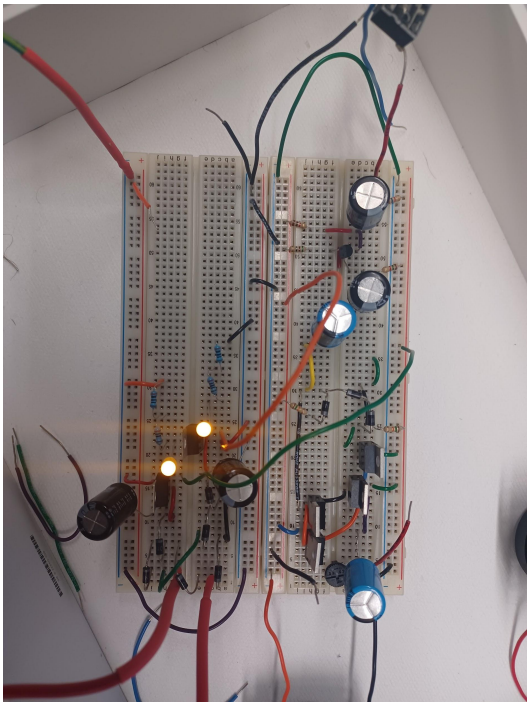


Figure 4.3. Implementation of the entire circuit on breadboard

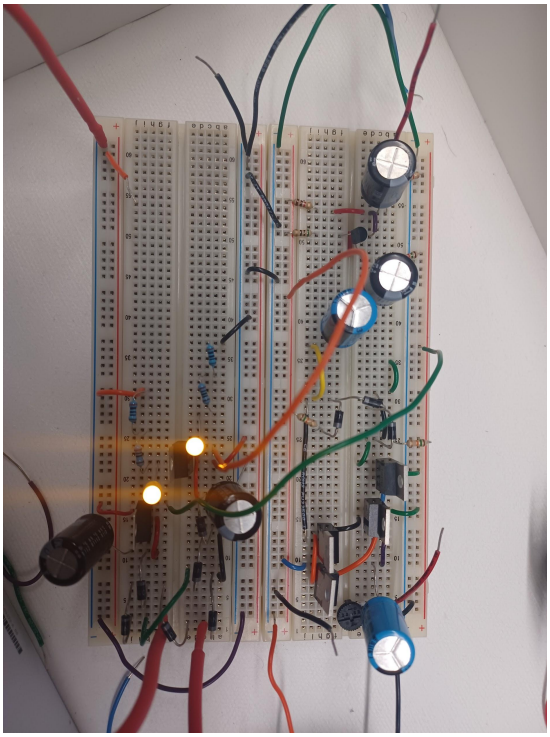


Figure 4.4. Implementation of the entire circuit on breadboard

6. *Discussion and Analysis*

The input impedance of the CE was approximately 960 ohm and output impedance of 1.4 k ohm. The input impedance of the CC amplifier was about 14 k ohm. This was the value that we were expecting because we want the majority of the output voltage from CE to be dropped across the CC amplifier. The biasing voltages were closer to what we calculated, and had about 5-6 percent error. The output voltage swing for each stage was almost between -10 V to 10 V.

The simulated results obtained for the entire circuit were similar to those of the measured. The simulated circuit could not be hooked up to an “iPod” so there is no comparison between the circuits for music being played. But the overall voltage gain, input and output voltages can be compared. For the input voltage of the overall simulated circuit the student set it to 495mVpk. The recorded input voltage for the simulated circuit was found to be -494mVpk and the output voltage was found to be 8.81V. With these two values the student found the overall voltage gain of the simulated circuit to be -17.83V/V. This value is only 12% different from that of the measured values of the circuit. This percent error is very good because it shows the student that the “real” circuit worked correctly with little to no errors. It is usually hard to get your circuit to work just as well as your simulated one because the simulated circuit is the perfect ideal situation and in the real world there is no “perfect” circuit.

The measured values of the Common-Emitter were what the student expected. The values followed that of what a Common-Emitter is supposed to do which is have a moderate input and output impedance and a large voltage gain. The input impedance of the common emitter was found to be 958Ω and the output impedance was found to be 1.4kΩ. The input impedance of the common-emitter is meant to be a value that is close to/matches the input impedances of the instruments being used with it. The moderate high output impedance allows for there to be a large gain on the amplifier circuit. We cannot just attach the speaker directly to the common emitter because when a low load is added to the amplifier (such as the speaker), this load decreases the gain of the amplifier. For the common-emitter amplifier to have a load attached and not have the gain affected the load must be very high and almost equal to that of the output impedance. The student measured the input and output voltages of the common-emitter using the oscilloscope for verification. The input voltage was measured to be 495mV and the output voltage was measured to be 10.6V. This gives the common- emitter an overall voltage gain of 21.41V/V. This voltage gain is 1V/V off of the entire voltage gain of the circuit which makes sense because the overall voltage gain of the common collector amplifier is 1V/V. This voltage gain is a great voltage gain because it follows exactly what the voltage gain of a common-emitter is to have.

The student decided to use the Darlington Pair Configuration in this final experiment

because just one transistor in the common-collector was not producing the output power needed for this project. The Darlington Pair Configuration allowed the student to reach the desired output power for the project. The reason the two 15kΩ biasing resistors were placed in the common collector circuit is because these two resistors biased the current that was going through the 4 Transistors so that only the right amount of current would go through them. The biasing resistors also help keep the current through the power supplies below 1 ampere.

7. Conclusion

In conclusion, both stages provided the expected output which was to first give a good quality of sound and stabilized thermal across the transistors. The Voltage gain had a lower percent error which was about 7 to 8 percent. However, the current gain between simulation and calculation was really high. The overall power gain was not as we expected to be due the fact that it was almost unattainable to have an output power of 10 Wrms. However, our design's output power was not exaggeratedly off from the expected output power. For instance, our simulation showed an output power of 6.89 W which is great considering the limitations that we had.